Abhishek Tripathi

AI / GenAI Engineer | LLM Applications, RAG & Agentic Systems

Pune, India (Open to remote roles across India) • +91 7668617782 • abhishek0akt@gmail.com • LinkedIn • GitHub

PROFESSIONAL SUMMARY

Data Scientist with **2+ years** building **production Generative AI and machine learning systems**: RAG pipelines, LLM applications, **multi-agent assistants** and predictive models for **enterprise clients across manufacturing, finance and HR tech**. Owns delivery **end to end**: problem framing, EDA, **LLM orchestration (LangChain, LangGraph)**, evaluation harnesses and cloud deployment on **Azure OpenAI and AWS**. **Published researcher** in deep learning computer vision (ETASR, 2024); **mentored a 4–5 person delivery pod**.

PROFESSIONAL EXPERIENCE

ASSOCIATE DATA SCIENTIST | *AlgoAnalytics Pvt. Ltd.* | *Jun 2025 – Present*

Agentic AI & Applied GenAI Products

- **Surveillance Suite Support Chatbot:** Designed and built a **multi-agent AI assistant** for the **microservice-based Surveillance Suite 2.1**, letting operators ask plain-language questions about cameras, alerts, live status, insight reports and troubleshooting in **English, Hindi, Marathi and Japanese**. A supervisor router, forced into **one structured tool call**, picks one of **12 specialised agents** (camera operations, alert investigation, live monitoring, KPI reports, system health and more); answers are written by an OpenAI model.
- **Surveillance Suite – single-Lambda, secure architecture:** A **single FastAPI + LangGraph deployment on AWS Lambda** (API Gateway, response streaming) serves **both platform versions** and is **plug-and-play for the mobile app**, with **no database access to customer data**. An **Edge Connector** exposes **14 allow-listed, read-only operations over AWS IoT Core MQTT**, so cameras, databases and internal APIs are never exposed to the internet. Every turn persists in **DynamoDB** so reloads and dropped connections recover the answer; API keys, usage limits, a daily cost reminder and re-runnable deployment scripts protect operations.
- **Surveillance Suite – product features & validation:** Built a **React chat widget** with evidence images, live camera view and docked and floating panels, plus an **"AI Insights" overview** with an activity heatmap drawn on a real camera frame. Validated the agents with an automated **57-question scoring run** and a **74-check end-to-end browser suite**.
- **Plant Live – unified operations dashboard:** Built an interactive prototype for a **multi-plant automotive sheet-metal components manufacturer (3 plants across India)** that pulls **10 AI and analytics systems** (including Surveillance Suite video analytics, aMIRAI shop-floor tickets, aQuality inspection, InsightEdge rejection analytics, MTC checks, a cutting optimizer and invoice/GST reconciliation) into **one live view**, centred on a **Three.js 3D digital twin** of each plant with animated machines, workers, forklifts and ANPR-tracked trucks, and roofs recoloured by safety, OEE or staffing.
- **Plant Live – decision support:** Delivered live KPIs, **camera-linked safety alerts** with response deadlines and automatic escalation, a **six-stage order tracker** and drill-downs for shops, inspections, control charts and cutting layouts. An **Executive mode** generates a rule-based daily brief (attention items, billing at risk, cross-plant scorecard, seven-day trends), and an **"Ask about today" LLM assistant** answers questions grounded in live plant data; responsive light/dark UI with resizable panels and keyboard support.

GenAI Platforms, RAG & Evaluation

- **AlgoCreator:** Reduced manual presentation effort **~70% (≈45 analyst hours/month)** and lifted **retrieval accuracy 35%** on a 100-query evaluation set by architecting an end-to-end **GenAI product generating branded PPTX decks from 1,000+ slides** through an **OpenAI GPT RAG pipeline on LangChain and ChromaDB**, with re-engineered chunking, embedding and multi-query search strategies.
- **AI-based recruiting:** Cut **recruiter screening time 60%** across **2 client accounts** and multiple hiring sprints by architecting a platform on **Azure OpenAI and Azure AI Search**, automating JD-to-candidate matching, **AI-avatar interviews** and post-interview scorecards.
- **Chatbot with AI insights:** Delivered **production RAG document-intelligence chatbots for 5 enterprise clients at 90%+ answer accuracy** by building ingestion, validation and transformation pipelines over multi-source structured and unstructured data on **AWS and Azure AI**.
- **Evaluation & regression tooling:** Standardised **quality gates across GenAI deliveries** with custom harnesses that run against live backends (a **100-query retrieval set** for AlgoCreator, a **30-query golden suite** for agent routing), backed by **structured per-turn logging** (thread and turn IDs) that makes every regression **reproducible and traceable to a single request**.

Predictive Modelling & Delivery Leadership

- **OneStopAI – predictive modelling:** Selected the **production churn model** by benchmarking Logistic Regression, **XGBoost** and gradient-boosted ensembles under **stratified k-fold cross-validation** on manufacturing and finance datasets, then extended the winner into a **personalised recommendation engine** surfaced in Tableau and Power BI.
- **OneStopAI – delivery leadership: Mentored 4–5 interns through 6+ production ML use cases** across manufacturing and finance, owning problem framing, code review and delivery standards from EDA through dashboard handover.

DATA SCIENCE CONSULTANT* | *AlgoAnalytics Pvt. Ltd.* | *Jan 2025 – May 2025

- **Japanese document extraction:** Automated processing of **500+ Japanese-language .docx/.xlsx documents per batch**, saving **15–20 hours of manual review**, by building a **PaddleOCR + LangChain + OpenAI GPT** extraction pipeline deployed as a **containerized FastAPI microservice (Docker) on AWS App Runner**.
- **Semantic slide retrieval:** Reduced slide search time **from hours to seconds** by developing **ChromaDB multi-query semantic retrieval** across thousands of client presentations with PDF/PPTX export, kept reproducible through Git version control and technical documentation.

***ASSISTANT DATA SCIENTIST (RESEARCH)* |** *MIT-WPU | Mar 2024 – Jul 2024*

- **Plant disease detection:** Achieved **95.93% precision and 88.38% mAP50-95** in real-time plant disease detection by training and benchmarking **YOLOv8n and YOLOv9c (PyTorch)** on precision, recall, F1 and mAP, then converted YOLOv8 to **TensorFlow Lite for Android deployment** used in live research trials.

CORE SKILLS & TECHNOLOGIES

GenAI & LLMs: OpenAI GPT, Azure OpenAI, Gemini Pro, LangChain, LangGraph, RAG pipelines, multi-agent orchestration, tool / function calling, prompt engineering, embeddings, chunking, semantic & multi-query search, evaluation harnesses, vector databases (ChromaDB, FAISS, Pinecone)
ML & Statistics: Python, SQL, R, Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow Lite, XGBoost, ensembles, neural networks, computer vision (YOLOv8/v9), NLP, feature engineering, hypothesis testing, A/B testing, EDA, forecasting, recommendation systems
Cloud & Deployment: AWS (Lambda, API Gateway, DynamoDB, IoT Core, S3, ECR, App Runner), Azure (OpenAI, AI Search, Document Intelligence), GCP Vertex AI, Docker, nginx, FastAPI, REST APIs, Git
Frontend: React, TypeScript, Vite, Three.js, Streamlit
Data & BI: MySQL, PostgreSQL, MongoDB, Redis, Kafka, Snowflake, ETL and data pipelines, Tableau, Power BI, Matplotlib, Seaborn

PUBLICATION

Comparative Analysis of YOLOv8 and YOLOv9 Models for Real-Time Plant Disease Detection in Hydroponics | ETASR, Vol. 14, No. 5, 2024, pp. 17269–17275 | DOI 10.48084/etasr.8301

EDUCATION

PG Diploma (AI & Machine Learning) | **MIT-WPU, Pune** | Aug 2023 – Jun 2024 | CGPA 9.0/10
B.Sc. (Mathematics: probability, statistics, mathematical modeling) | **CSJMU, Kanpur** | 2020 – 2023

CERTIFICATIONS

- **Career Essentials in Generative AI** – Microsoft & LinkedIn
- **Foundations: Data, Data, Everywhere** – Google
- **AWS Academy Graduate – Machine Learning Foundations** – AWS Academy